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Distributed Stochastic AC Optimal Power Flow based on Polynomial Chaos Expansion

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Abstract— **Distributed optimization methods for Optimal Power Flow (OPF) problems are of importance in reducing coordination complexity and ensuring economic grid operation. Renewable feed-ins and demands are intrinsically uncertain and often follow non-Gaussian distributions. The present paper combines uncertainty propagation via Polynomial Chaos Expansion (PCE) with the Augmented Lagrangian Alternating Direction Inexact Newton (ALADIN) method to solve stochastic OPF problems with non-Gaussian uncertainties in a distributed setting. Moreover, using ALADIN and PCE we obtain fast convergence while avoiding computationally expensive sampling. A numerical example illustrates the performance of the proposed approach.**

I. INTRODUCTION

The aim of Optimal Power Flow (OPF) is to compute set points for power injections in grids minimizing the overall operation cost while satisfying power balance, and meeting generation and line capacity limits. Often, OPF problems are solved for the high-voltage transmission part of grid relying on convex DC approximations. At the same time, the majority of renewable feed-in is located at (mid- to low-voltage) parts of the grid (i.e. at the distribution grid), where DC-like approximations are not sufficiently accurate and the number of grid nodes to be considered can become quite large. Thus, (i) there exists a need for development of efficient distributed algorithms for AC-OPF problems. Moreover, renewable feed-in and demand are intrinsically uncertain. Hence, (ii) the consideration of stochastic uncertainties in OPF problems is of particular interest.

Concerning (i), several works directly apply methods from convex optimization like the Alternating Direction Method of Multipliers (ADMM) to the non-convex AC-OPF problem [1, 2]. In this case, there is usually no convergence guarantee and the convergence rate is often slow [3]. Other approaches use outer [4] or inner convex approximations [5]. In the former, the feasibility of the solution can be guaranteed for special cases like radial grids, but not for general meshed grids [6]. In the latter, inner approximations lead to conservative solutions [5]. For a recent overview on distributed OPF consider [7]. In contrast, in [8] we suggest applying the Augmented Lagrangian Alternating Direction Inexact Newton (ALADIN) method—which provides certain convergence guarantees for non-convex distributed problems [9]—to solve the non-convex AC-OPF problem without any approximations.

Concerning (ii), it is worth noting that the consideration of stochastic uncertainties in OPF problems has recently become an active area of research [10, 11]. Typically, in stochastic OPF one computes solutions parameterized as functions of the uncertainty that can be used for different purposes: One application is to use the resulting probability density functions for planning of reserve capacities. Another possibility is to use the resulting parameterized solutions directly as control laws, see [10, 12].

An important aspect in stochastic OPF is the reformulation of inequality constraints. For example, joint chance constraint formulations that can be solved by means of scenario-based optimization techniques have been proposed [13]. Similarly, individual chance constraint reformulations for the OPF problem in conjunction with deterministic convex reformulations have shown promising results [12, 14, 15]. However, the above works consider the simplified DC setting. There exists much fewer results for non-convex AC-OPF [10, 16, 17]. Moreover, besides chance constraint reformulations for the inequality constraints, also the equality constraints stemming from the power flow equations are of specific interest in stochastic OPF problems. Specifically, one wants to ensure their satisfaction for any realization of the uncertainty, coined *viability* in [12]. For the DC setting viability can be established [12, 15], while a detailed analysis of the AC case is still missing. Also, the considered class of uncertainties is often restricted to Gaussian distributions [12, 14]. However, it has been pointed out in the literature that non-Gaussian random variables may be more appropriate than Gaussians when modeling load patterns or renewable energy production [18, 19].

Under mild technical assumptions, Polynomial Chaos Expansion (PCE) allows considering Gaussian and non-Gaussian uncertainties alike in a unified framework [20, 21]. Recently, PCE has seen several applications to control and optimization problems [22]. In the context of power systems, PCE has been applied to centralized DC-OPF problems [15] and to centralized AC-OPF problem [16].

The contribution of the present paper is to combine both lines of research. We investigate algorithms for solving

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stochastic AC-OPF problems in a distributed setting, which are based on ALADIN [9]. Our results indicate that ALADIN is well-suited for solving distributed stochastic OPF problems, and that even low-order PCE solutions lead to accurate approximations of the power-flow equations. The remainder of the paper is structured as follows: Section II provides a concise problem statement. Section III reviews ALADIN. Section IV presents numerical results.

Notation: While $x \in \mathbb{R}^n$ is a deterministic vector, $\mathsf{x} \in L^2(\Omega, \mu; \mathbb{R}^n)$ denotes the corresponding real-valued random vector with sample space Ω, and probability measure μ. $X \in \mathbb{R}^{n \times L+1}$ the corresponding matrix of PCE coefficients. The inner product of functions $\phi_k : \Omega \to \mathbb{R}$ and $\phi_l : \Omega \to \mathbb{R}$ with respect to the measure $\mu : \Omega \to [0, 1]$ is defined as $\langle \phi_k, \phi_l \rangle := \int_\Omega \phi_k(s)\phi_l(s)\,\mathrm{d}\mu(s) = \int_\Omega \phi_k(s)\phi_l(s)\rho(s)\,\mathrm{d}s$ and $\rho : \Omega \to \mathbb{R}^+$ is the corresponding probability density function (PDF). Expected value and variance of a random variable $\mathsf{x} \in L^2(\Omega, \mu; \mathbb{R})$ are denoted as $\mathbf{E}[\mathsf{x}] = \int_\Omega t\rho(t)\mathrm{d}t$ and $\mathbf{Var}[\mathsf{x}] = \mathbf{E}\left[(\mathsf{x} - \mathbf{E}[\mathsf{x}])^2\right]$ respectively.

II. Preliminaries & Problem Formulation

Starting from the deterministic OPF problem in separable form, this Section states the OPF problem in terms of random variables, and then derives a deterministic reformulation thereof in terms of spectral expansion coefficients.

The considered electrical grid at steady state $(\mathcal{N}, \mathcal{G}, \mathcal{L}, \mathcal{W})$ is described by a node set $\mathcal{N} = \{1, \ldots, N\}$, a generator set $\mathcal{G} \subseteq \mathcal{N}$, a line set $\mathcal{L} \subseteq \mathcal{N} \times \mathcal{N}$, and transmission line parameters $\mathcal{W} = \{(g_{kl}, b_{kl}) \,|\, (k,l) \in \mathcal{L}\}$, where g_{kl} and b_{kl} are the line conductances and susceptances. The node set $\mathcal{N}$ is partitioned into $\mathcal{R} = \{1, \ldots, R\}$ regional nodes sets $\mathcal{N}_i = \{n_1^i, \ldots, n_{N_i}^i\} \subset \mathcal{N}$, $i \in \mathcal{R}$ with $\cup_{i \in \mathcal{R}} \mathcal{N}_i = \mathcal{N}$ and $\mathcal{N}_i \cap \mathcal{N}_j = \emptyset$ for $i \neq j$. A set of auxilliary node pairs $\mathcal{A} \subset \mathcal{N} \times \mathcal{N}$ collects all nodes located at borders between two neighboring regions [23].[1] Fig. 1 shows an example for grid partitioning using the introduced notation.

Every node $k \in \mathcal{N}$ is described by four physical quantities, $\psi_k = \begin{bmatrix} v_k^{\text{re}} & v_k^{\text{im}} & p_k & q_k \end{bmatrix}^\top \in \mathbb{R}^4$, where v_k^{re} is the real part and v_k^{im} is the imaginary part of the voltage phasor at node k. Moreover, p_k and q_k denote the active and reactive power injection at node k. All nodal variables ψ_k from region i are collected in the vector of local/regional variables, i.e. $x_i = [\psi_{n_1^i}{}^\top, \ldots, \psi_{n_{N_i}^i}^\top]^\top \in \mathbb{R}^{n_i}$ with $n_i = 4N_i$ for all regions $i \in \mathcal{R}$. Active and reactive power demands at node k, denoted by $p_k^{\text{nd}} \in \mathbb{R}$ and $q_k^{\text{nd}} \in \mathbb{R}$, are written as local demand vectors $p_i^d = [p_{n_1^i}^{\text{nd}}, \ldots, p_{n_{N_i}^i}^{\text{nd}}]^\top$ and $q_i^d = [q_{n_1^i}^{\text{nd}}, \ldots, q_{n_{N_i}^i}^{\text{nd}}]^\top$, for all regions $i \in \mathcal{R}$. To simplify notation the local demand vectors p_i^d and q_i^d are subsumed as local parameter vectors $w_i = [\, p_i^{d\top},\ q_i^{d\top}\,]^\top \in \mathbb{R}^{2N_i}$.

[1] Due to space limitations, we assume that the auxiliary nodes are already introduced and added to the node set $\mathcal{N}$, cf. [23] for details. By doing so, the original node set is enlarged e.g. from 5 to 13 nodes for the IEEE 5–bus case shown in Fig. 1.

1) Separable AC-OPF *Problem:* As in [8, 9], we consider the AC-OPF problem in affine-coupled separable form

$$\min_{x \in \mathbb{R}^{n_x}} \quad \sum_{i \in \mathcal{R}} f_i(x_i) \tag{1a}$$

$$\text{s.t.} \quad h_i(x_i, w_i) = 0 \qquad \forall\, i \in \mathcal{R} \tag{1b}$$

$$x_i \in [\underline{x}_i,\ \bar{x}_i] \qquad \forall\, i \in \mathcal{R} \tag{1c}$$

$$\sum_{i \in \mathcal{R}} A_i x_i = 0, \tag{1d}$$

where $x = [x_1^\top, \ldots, x_R^\top]^\top \in \mathbb{R}^{n_x}$. In the following we briefly recap how to set up the components of (1), see e.g. [8] for details.

For all regions $i \in \mathcal{R}$, the local objective functions $f_i : \mathbb{R}^{n_i} \to \mathbb{R}$ from (1a) are

$$f_i(x_i) := \sum_{k \in \mathcal{G}_i} c_{1,k} p_k^2 + c_{2,k} p_k + c_{3,k}, \quad c_{1,k} > 0, \tag{2}$$

with local generator sets $\mathcal{G}_i = \mathcal{N}_i \cap \mathcal{G}$. The power balance in an AC grid in steady state is modeled by the power flow equations. In rectangular coordinates, for all $k \in \mathcal{N}_i$ and $i \in \mathcal{R}$, they read [24]

$$\begin{aligned} p_k^{\text{nd}} - p_k &= \sum_{l \in \mathcal{N}_i} g_{kl}(v_k^{\text{re}} v_l^{\text{re}} + v_k^{\text{im}} v_l^{\text{im}}) + b_{kl}(v_k^{\text{im}} v_l^{\text{re}} - v_k^{\text{re}} v_l^{\text{im}}), \\ q_k^{\text{nd}} - q_k &= \sum_{l \in \mathcal{N}_i} g_{kl}(v_k^{\text{im}} v_l^{\text{re}} - v_k^{\text{re}} v_l^{\text{im}}) - b_{kl}(v_k^{\text{im}} v_l^{\text{im}} + v_k^{\text{re}} v_l^{\text{re}}), \end{aligned} \tag{3}$$

where $g_{kk} = \sum_{l \in \mathcal{N}_i \setminus \{k\}} -g_{kl}$ and $b_{kk} = \sum_{l \in \mathcal{N}_i \setminus \{k\}} -b_{kl}$. The power flow equations (3) are the major source of nonlinearity and nonconvexity in (1b). A reference voltage phasor is required to obtain a unique solution of the power flow equations. Hence, in one region $i \in \mathcal{R}$, we introduce a reference node $r \in \mathcal{N}_i$, cf. [24], where we enforce $v_r^{\text{re}} = 1$, $v_r^{\text{im}} = 0$. This constraint together with the power flow equations (3) constitute local equality constraints $h_i : \mathbb{R}^{n_i} \times \mathbb{R}^{2N_i} \to \mathbb{R}^{n_{hi}}$ from (1b). The box constraints in (1c) collect bounds for active and reactive power injections p_k, q_k, and the real parts of the voltages v_k^{re} for all nodes $k \in \mathcal{N}$. For simplified presentation only the real parts of the voltage are constrained. This is valid only for small phase angles. A more rigorous approach would be to constrain the voltage magnitude.

At all auxiliary node pairs $(k,l) \in \mathcal{A}$, the couplings between two neighboring regions are described by $v_k^{\text{re}} = v_l^{\text{re}}$, $p_k = -p_l$, $v_k^{\text{im}} = v_l^{\text{im}}$ and $q_k = -q_l$. This yields the consensus constraint (1d), where A is constructed such that the above equations are satisfied for all $(k,l) \in \mathcal{A}$.

Problem (1) is parameterized by the local power demands w_i. However, the increasing renewable generation units calls for the consideration of *uncertain* and *stochastic* power demand/generation.

2) Random Variable OPF Problem: In the following, we describe how to reformulate Problem (1) in terms of random variables. The resulting "optimal random variables"—which are essentially L^2 functions—can afterwards be used in context of planning of reserve capacities or as control laws avoiding computationally expensive sampling, cf. Section I.

The necessity to model power demand/generation as random variables has been pointed out e.g. by [18, 19]. In Problem (1b) we replace the parameters w_i with random variables $\mathsf{w}_i \in L^2(\Omega, \mu; \mathbb{R}^{2N_i})$ with corresponding sample space Ω, and probability measure μ [25]. Doing so, the objective function f and the constraint functions h_i and box constraints (1c) turn into functions mapping random variables to random variables. Importantly, the decision variable becomes a random variable too. Reformulating the deterministic Problem (1), we obtain the following random variable OPF problem

$$\min_{\mathsf{x} \in L^2(\Omega,\mu;\mathbb{R}^{n_x})} \quad \sum_{i\in\mathcal{R}} \mathbf{E}[f_i(\mathsf{x}_i)] \tag{4a}$$

$$\text{s.t.} \quad \forall\, i \in \mathcal{R}:$$

$$h_i(\mathsf{x}_i, \mathsf{w}_i) = 0 \tag{4b}$$

$$\mathbf{E}[\mathsf{x}_{i,j}] \pm \beta_{i,j}\sqrt{\mathbf{Var}[\mathsf{x}_{i,j}]} \in [\underline{x}_{i,j},\ \bar{x}_{i,j}] \quad \forall\, j \in \mathcal{J}_i \tag{4c}$$

$$\sum_{i\in\mathcal{R}} A_i \mathsf{x}_i = 0, \tag{4d}$$

where $\mathsf{x} = [\mathsf{x}_1^\top, \dots, \mathsf{x}_R^\top]^\top \in L^2(\Omega, \mu; \mathbb{R}^{n_x})$ and $\mathcal{J}_i := \{1, \dots, n_i\}$. Now, we minimize the expected value of the objective function instead of the objective itself.[2] Note that all equality constraints—power flow equations, slack constraints, and consensus constraints—are written in terms of random variables [15, 16]. For the scope of this paper we assume that the power flow equations admit a physically meaningful solution for all realizations w_i of w_i.[3]

Remark 1 (Viability): The power flow equations have to hold for every possible realization of the uncertainty. This notion is called *viability* in [12]. Writing the power flow equations in terms of random variables means that for every realization of the uncertainty a realization of the power generation has to be attained. We remark that a reformulation in terms of chance constraints would be not physically relevant and is thus not pursued here. □

Inequality constraints in terms of random variables are, excluding special/pathological cases, not directly meaningful. Hence, they are reformulated as individual chance constraints

$$\mathbf{P}\left[\mathsf{x}_{i,j} \geq \underline{x}_{i,j}\right] \geq 1 - \epsilon_{i,j},\ \mathbf{P}\left[\mathsf{x}_{i,j} \leq \bar{x}_{i,j}\right] \geq 1 - \epsilon_{i,j},$$

for $i \in \mathcal{R}$, which are reformulated deterministically as

$$\mathbf{E}[\mathsf{x}_{i,j}] \pm \beta_{i,j}(\epsilon_{i,j})\sqrt{\mathbf{Var}[\mathsf{x}_{i,j}]} \in [\underline{x}_{i,j}\ \ \bar{x}_{i,j}], \tag{5}$$

for all $j \in \mathcal{J}_i$ and form (4c) [12, 14]. For example, if x_i is Gaussian and $\beta_{i,j}(\epsilon_{i,j}) = \Psi^{-1}(1 - \epsilon_{i,j})$ holds, where Ψ is the cumulative distribution function of the standard normal distribution, then the reformulation (5) is exact. For arbitrary distributions (with known moments), the choice of $\beta_{i,j}(\epsilon_{i,j})$ can ensure distributional robustness [14]. In contrast to (1), Problem (4) is infinite-dimensional due to the decision variables x_i being random variables, i.e. elements of an L^2-space.

Remark 2 (Optimal Random Variable): Assuming the solution to Problem (4) exists it corresponds to an optimal random variable $\mathsf{x}^\star$. Mathematically, the optimal solution $\mathsf{x}^\star$ corresponds to an L^2 function in terms of a standard random variable, which can be interpreted as a policy. That means a given the realization of the standard random variable leads to a realization $x^\star$ of the optimal solution $\mathsf{x}^\star$. This specific realization satisfies the power flow equations; the inequality constraints are valid in the chance-constraint sense, i.e. they hold with a user-specified probability. □

3) Finite Parameterization of Stochastic AC-OPF*:* Next, similar to other optimization and optimal control problems in function spaces, we reformulate Problem (4) as a finite-dimensional optimization problem using spectral expansion. By introducing Polynomial Chaos Expansion to Problem (4)—a spectral expansion technique for random variables of finite variance—we can express Problem (4) in terms of deterministic PCE coefficients. A thorough introduction of PCE is beyond the scope of this paper; for details we refer to [20, 21].

Consider a vector of independent standard random variables $\mathsf{s} \in L^2(\Omega, \mu; \mathbb{R}^{n_s})$, and polynomials $\phi_l : \mathbb{R}^{n_s} \to \mathbb{R}$, $l = 0, \dots, L$ that are orthogonal with respect to the measure of s, i.e. $\langle \phi_i, \phi_j \rangle = \alpha_i \delta_{ij}$, where δ_{ij} is the Kronecker-delta and $\alpha_i = \langle \phi_i, \phi_i \rangle \in \mathbb{R}$ is positive. Then, the truncated polynomial chaos expansion of the random vector x_i is

$$\mathsf{x}_i(\mathsf{s}) = \sum_{l=0}^{L} \chi_{l,i}\phi_l(\mathsf{s}) = X_i\, \Phi(\mathsf{s}), \tag{6}$$

with $X_i = [\chi_{0,i}, \dots, \chi_{L,i}]$ and $\Phi = [\phi_0, \dots, \phi_L]^\top$. The vectors $\chi_{l,i} \in \mathbb{R}^L$ are called PCE coefficients.[4] For the parameter vectors w_i PCEs are given analogously to (6) with PCE coefficients $\omega_{l,i} \in \mathbb{R}^{n_{wi}}$ and a corresponding coefficient matrix $W_i = [\omega_{0,i}, \dots, \omega_{L,i}]$.[5]

The basis polynomials ϕ_l are chosen based on the underlying probability density function of the random variables s_i according to the Askey scheme [30]. For example, for random variables that follow a Gaussian/Beta/Gamma/Uniform distribution the respective orthogonal polynomials are known to be Hermite/Jacobi/Laguerre/Legendre [20, 21]. The dimension n_s represents the number of independent "sources of uncertainty", e.g. in case of one independent wind feed-in and one demand, n_s would be 2.

[2]We remark that this is a modeling choice. Other reformulations exist, e.g. minimize a combination of objective value and variance.

[3]Note that determining the feasibility of power flow equations for a given parametrization is in general difficult [26–28].

[4]In general $L = \infty$ is needed for an exact PCE. However, recently it has been shown that under certain assumptions the truncation error vanishes or can be quantified [29] precisely.

[5]In order to render PCE applicable, one needs to map realizations of the uncertain demands w to s. Note that this does not pose problems whenever the PCE representation of the uncertainty is affine and the dimension of s is smaller than the number of measured uncertainties.

$$
\begin{aligned}
(P_i - P_{i,d})\langle \Phi, \phi_k\rangle &= \sum_{j=1}^{N} g_{ij}\left(V_i^{\mathrm{re}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{re}\top} + V_i^{\mathrm{im}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{im}\top}\right) + b_{ij}\left(V_i^{\mathrm{im}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{re}\top} - V_i^{\mathrm{re}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{im}\top}\right)\\
(Q_i - Q_{i,d})\langle \Phi, \phi_k\rangle &= \sum_{j=1}^{N} g_{ij}\left(V_i^{\mathrm{im}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{re}\top} - V_i^{\mathrm{re}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{im}\top}\right) - b_{ij}\left(V_i^{\mathrm{im}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{im}\top} + V_i^{\mathrm{re}}\langle \Phi\Phi^\top\!, \phi_k\rangle V_j^{\mathrm{re}\top}\right)
\end{aligned} \tag{7}
$$

By introducing PCE, (4) becomes

$$
\min_{X_1,\dots,X_R} \quad \sum_{i\in\mathcal{R}} \mathbf{E}[f_i(X_i\,\Phi(\mathsf{s}))] \tag{8a}
$$

$$
\text{s.t.} \quad \forall\, i \in \mathcal{R}:
$$

$$
h_i(X_i\,\Phi(\mathsf{s}), W_i\,\Phi(\mathsf{s})) = 0 \tag{8b}
$$

$$
\mathbf{E}[X_{i,j}\,\Phi(\mathsf{s})] \pm \beta_{i,j}\sqrt{\mathbf{Var}[X_{i,j}\,\Phi(\mathsf{s})]} \in [\underline{x}_{i,j}\;\bar{x}_{i,j}] \quad \forall j \in \mathcal{J}_i \tag{8c}
$$

$$
\sum_{i\in\mathcal{R}} A_i X_i\,\Phi(\mathsf{s}) = 0, \tag{8d}
$$

where the PCE coefficients W_i are assumed to be given and $X_{i,j}$ denotes the j-th row of X_i. Observe that the decision variables in (8) are now deterministic PCE coefficients.

Problem (8) is in general infeasible for $L < \infty$, since the right hand side of the power flow equations (3) considered in (8b) is quadratic while the left hand side is linear in x_i. Hence, we apply Galerkin projection to (3) and reference constraints, in order to approximately consider these equations in the finite dimensional subspace spanned by $\{\phi_0, \dots, \phi_L\}$. We will show in the simulations, that this approach leads to acceptably small errors. Equation (7) shows the Galerkin projected power flow equations. Observe that (7) contains only PCE coefficients, i.e. the dependence on s is eliminated, cf. [16] for detailed derivation. Hereby, capitalized letters for the physical values (e.g. $V_i^{\mathrm{re}} \in \mathbb{R}^{L+1}$ for $v_i^{\mathrm{re}} \in \mathbb{R}$) denote row vectors of PCE coefficients for the the respective physical value. The vectors of scalar products,

$$
\langle \Phi, \phi_l\rangle := \langle \phi_l, \phi_l\rangle\, e_l, \tag{9}
$$

where e_l is the l-th unit vector, and the matrices

$$
\langle \Phi\Phi^\top\!, \phi_l\rangle := \begin{bmatrix} \langle \phi_1\phi_1, \phi_l\rangle & \dots & \langle \phi_1\phi_L, \phi_l\rangle \\ \vdots & \ddots & \vdots \\ \langle \phi_L\phi_1, \phi_l\rangle & \dots & \langle \phi_L\phi_L, \phi_l\rangle \end{bmatrix}, \tag{10}
$$

for $l = 1, \dots, L$ are fixed and can be precomputed. The reference constraints become $V_{r,1}^{\mathrm{re}} = 1$ and $V_{r,l}^{\mathrm{re}} = 0$ for $l = 1, \dots, L$ and $V_{r,l}^{\mathrm{im}} = 0$ for $l = 0, \dots, L$.

In order to eliminate the dependence on $\Phi(\mathsf{s})$ also in the objective function (8a) and box constraints (8c), we evaluate the expected value and variance. Thus, we define $F_i(X_i) := \mathbf{E}[f_i(X_i\,\Phi(\mathsf{s}))]$ where

$$
\mathbf{E}[f_i(X_i\,\Phi(\mathsf{s}))] = \sum_{j\in\mathcal{G}_i} c_{1,j}\mathbf{E}[\mathsf{p}_j^2] + c_{2,j}\mathbf{E}[\mathsf{p}_j] + c_{3,j} \tag{11}
$$

and $\mathbf{E}[\mathsf{p}_j] = P_{j,0}$ and $\mathbf{E}[\mathsf{p}_j^2] = \sum_{l=1}^{L} P_{j,l}^2\langle \phi_l, \phi_l\rangle$, as we assume $\phi_0(s) = 1$ without loss of generality, cf. [29].

Analogously, the inequality constraints (8c) are reformulated as

$$
\chi_{0,i,j} \pm \beta_{i,j}\sqrt{\textstyle\sum_{l=1}^{L} \chi_{l,i,j}^2\,\langle \phi_l, \phi_l\rangle} \in [\underline{x}_{i,j}\;\;\bar{x}_{i,j}] \tag{12}
$$

with $\mathbf{E}[\mathsf{x}_{i,j}] = \chi_{0,i,j}$, and $\mathbf{Var}[\mathsf{x}_i] = \sum_{l=1}^{L} \chi_{l,i,j}^2\,\langle \phi_l, \phi_l\rangle$, for all $i \in \mathcal{R}$. Moreover, applying Galerkin projection to the consensus constraint (8d) yields $\sum_{i\in\mathcal{R}} A_i\,\chi_{l,i} = 0$ for $l = 0, \dots, L$.

By combining the above, we obtain the following separable affine coupled, deterministic reformulation of (8)

$$
\min_{X_1,\dots,X_R} \quad \sum_{i\in\mathcal{R}} F_i(X_i) \tag{13a}
$$

$$
\text{s.t.} \quad H_i(X_i, W_i) = 0 \qquad \forall\, i \in \mathcal{R} \tag{13b}
$$

$$
G_i(X_i) \le 0 \qquad \forall\, i \in \mathcal{R} \tag{13c}
$$

$$
\sum_{i\in\mathcal{R}} A_i X_i = 0. \tag{13d}
$$

where F_i is given by (11), H_i collects the projected power flow equations (7) and reference constraints, G_i is given by (12) and the consensus constraint (13d) is given by the consensus constraint from above in matrix notation.

To be able to apply ALADIN to (13), from here on we treat X_i as the vectorization thereof, i.e. $X_i \leftarrow \mathrm{vec}(X_i)$ which results in vector-valued functions F_i, H_i, G_i and consensus constraint.

III. Solution via ALADIN

We suggest solving Problem (13) via the ALADIN method shown in Algorithm 1. For sake of concise notation, we treat the equality constraints (13b) as inequality constraints being always active. Thus, $\tilde{G} \le 0$ in ALADIN refers to the equality and inequality constraints (13b) and (13c). The algorithm is introduced very briefly, for a more detailed description we refer to [9]. In Step 1) of Algorithm 1, local augmented Lagrangians for fixed Lagrange multipliers λ^k and solution estimates Z^k subject to the local nonlinear constraints $\tilde{G}_i$ for all regions $i \in \mathcal{R}$ are minimized in parallel. Step 2) computes local approximations of the gradients g_i^k, the Hessians B_i^k and the Jacobians C_i^{*k} of the active nonlinear constraints $\tilde{G}_i$ and communicates them to a central entity. Step 3) solves an equality constrained quadratic program composed of the derivatives from the previous step with $B^k := \mathrm{diag}(B_1^k, \dots, B_R^k)$, $g^k := [g_1^{k\top}, \dots, g_R^{k\top}]^\top$ and $C^{*k} := \mathrm{diag}(C_1^{*k}, \dots, C_R^{*k})$. After a line search Step 4)—that from our experience can often be omitted in practice—the Lagrange multipliers and solution estimates are updated in Step 5). Finally, the updated values are broadcasted to the local entities and the next iteration starts. For a detailed discussion of convergence properties of ALADIN see [9].

Result: $X^\star$
Input: $Z^0,\ \lambda^0,\ \rho,\ \Sigma_i,\ \mu,\quad k=0,\ X^0=\infty;$

while $\|AX\|_1 > \epsilon$ ***and*** $\rho\,\|\Sigma_i(X_i - Z_i)\|_1 > \epsilon$ **do**

1) Solve local problems

$$X_i^k = \underset{X_i}{\operatorname{argmin}} \quad F_i(X_i) + \lambda^{k\top} A_i X_i + \frac{\rho}{2}\|X_i - Z_i^k\|^2_{\Sigma_i}$$
$$\text{s.t.} \qquad \tilde{G}_i(X_i) \le 0 \quad | \ \kappa_i^k$$

2) Compute local gradients & Hessians for QP

$$g_i^k = \nabla F_i(X_i^k)$$
$$B_i^k = \nabla^2 (F_i(X_i^k) + \kappa_i^{k\top} \tilde{G}_i(X_i^k))$$
$$C_{i,j}^{*k} = \begin{cases} \nabla(\tilde{G}_i(X_i^k))_j & \text{if } \ (\tilde{G}_i(X_i^k))_j = 0 \\ 0 & \text{otherwise} \end{cases}$$

3) Solve coordination QP

$$\min_{\Delta X, t} \ \frac{1}{2}\Delta X^\top B^k \Delta X + g^{k\top}\Delta X + \lambda^\top t + \frac{\mu}{2}\|t\|_2^2$$
$$\text{s.t.} \quad A(X^k + \Delta X) - t = 0 \quad | \ \lambda_{QP}^k$$
$$C^{*k}\Delta X = 0$$

4) Line search
Details in [9] $\rightarrow \alpha_1^k, \alpha_2^k, \alpha_3^k$,
otherwise $\alpha_1 = \alpha_2 = \alpha_3 = 1$

5) Update

$$Z^{k+1} \leftarrow Z^k + \alpha_1^k(X^k - Z^k) + \alpha_2^k \Delta X^k,$$
$$\lambda^{k+1} \leftarrow \lambda^k + \alpha_3^k(\lambda_{QP}^k - \lambda^k),$$
$$k \leftarrow k+1$$

end

Algorithm 1: ALADIN algorithm.

IV. Numerical Simulation

1) Test Case: The test case shown in Fig. 1 is based on the IEEE 5-bus grid [31]. We introduce uncertain demands at node 2, uniformly distributed between 4 p.u. and 5.6 p.u.; at node 4, Gaussian distributed with mean 1 p.u. and variance $(0.1\,\text{p.u.})^2$); and a fixed demand of 3 p.u. at node 3.[6] Similar to the deterministic setting in [8], the grid is decomposed into three regions $\mathcal{R} = \{1, 2, 3\}$ containing a generation center in the west and consumption centers in the east containing the uncertain demands. The line limits of the original test case are neglected. This test case is usually hard for distributed algorithms, since it is highly meshed and the generators in the generation center are cheap, such that a large share of the generated power has to be transferred between regions.

2) Numerical Results: The basis Φ is chosen based on the PDFs of the underlying uncertainties according to the Askey scheme [30], i.e. an Hermite basis for the Gaussian uncertainty and Legendre for the uniform uncertainty. In this particular case we yield $\Phi(\mathsf{s}) = \begin{bmatrix}1, & \mathsf{s}_1, & \mathsf{s}_2, & 1.5\mathsf{s}_1^2 - 0.5, & \mathsf{s}_1\mathsf{s}_2, & \mathsf{s}_2^2 - 1\end{bmatrix}^\top$, where $\mathsf{s} = [\mathsf{s}_1, \mathsf{s}_2]^\top$. After precomputing the inner products $\langle \Phi\Phi^\top, \phi_l\rangle$ and $\langle \Phi, \phi_l\rangle$ for $l = 1, \ldots, L$ from (9) and (10), ALADIN solves the coefficient Problem (13) in a distributed fashion. Fig. 2 shows the progress towards a minimizer $\|x^k - x^*\|_2$ and the KKT residual $r^k = \|\nabla F(X^k) + A^\top \lambda^k + \nabla \tilde{G}(X^k)\kappa^k\|_2$ over the iteration index k. ALADIN converges quadratically after around 5 iterations and terminates after around 20 iterations yielding optimal coefficients X_i for all regions $i \in \mathcal{R}$.[7] The parameters for ALADIN, the PCEs and the chance constraints are chosen to be $\rho = 10^3$, $\mu = 10^2 \cdot 1.5^k$, $\bar{\mu} = 10^6$, $L = 5$ and $\beta_{i,j} = 3$.

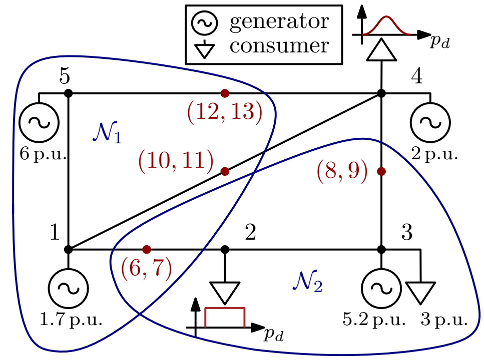

Fig. 1. Modified IEEE 5-bus test case with three regional node sets $\mathcal{N}_1 = \{1, 5, 6, 10, 12\}$, $\mathcal{N}_2 = \{2, 3, 7, 8\}$, $\mathcal{N}_3 = \{4, 9, 11, 13\}$ (blue), auxiliary node pairs $\mathcal{A} = \{(6, 7), (8, 9), (10, 11), (12, 13)\}$ (red).

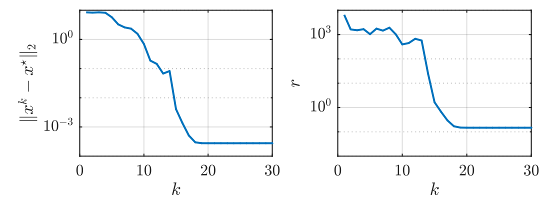

Fig. 2. Distance to the minimizer (left), and KKT residual (right).

Finally, we evaluate whether the generator bounds and the power flow equations are fulfilled. Therefore, we compute 10^5 realizations of the uncertainty s and evaluate the constraints for the corresponding realizations of the uncertain parameters w_i and decision variables x_i. The resulting histograms for the uncertain demands, selected feed-ins and the cost are shown in Fig. 3 and Fig. 4, where $\tilde{\rho}$ is the normalized frequency to the corresponding probability density function ρ. One can see that the cheapest generators hit their upper limits for the active power injections almost accurately. Hence, the reformulation of the constraints works reasonable for our case. Note that the accuracy of the constraint satisfaction depends on $\beta_{i,j}$.

Fig. 5 shows the Root-Mean-Square Error (RMSE) e of the violation of the power flow equations for the realizations of x and w. One can see that—despite the fact that the PCE is truncated after L terms—the constraint violation RMSE is smaller than 10^{-4} [p.u.] for all realizations of the uncertainty. The mean of the RMSE admits $2.35 \cdot 10^{-6}$ [p.u.] and the maximum RMSE is $22.8 \cdot 10^{-6}$ [p.u.].

[6] The coefficients for the reactive power demands are chosen to be 10% of the active power coefficients.

[7] In principle, ALADIN is able to solve optimization problems to a higher accuracy than 10^{-3}, cf. numerical results in [9]. The maximum accuracy of 10^{-3} reached here is caused by a prototypical implementation of ALADIN where all equality constraints are reformulated as two inequality constraints with a certain threshold, i.e. $-\alpha \le H_i(X_i, W_i) \le \alpha$ for all $i \in \mathcal{R}$ and $\alpha = 1 \cdot 10^{-6}$. Hence, the feasible set is enlarged and the solution obtained from ALADIN is slightly different to the "true" minimizer..

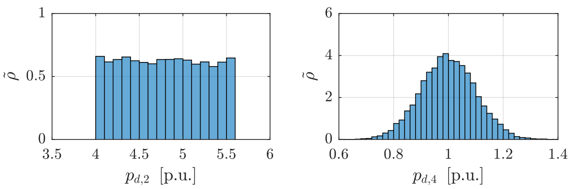

Fig. 3. Realizations of the parameter uncertainty.

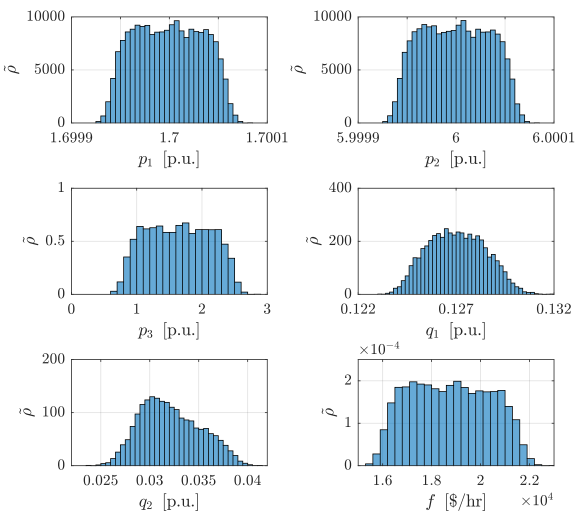

Fig. 4. Realizations of selected power feed-ins and the objective function.

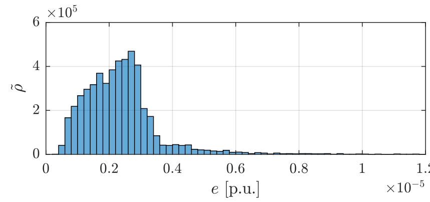

Fig. 5. Histogram of RMS error of the power flow equations (3).

V. Conclusion & Outlook

This paper investigates a first approach to distributed stochastic OPF considering non-Gaussian uncertainties. We use Polynomial Chaos Expansion to attain a deterministic reformulation of the stochastic OPF problem. Furthermore, the resulting separable problem is solved via ALADIN obtaining convergence guarantees and fast convergence. We show numerically, that despite the truncation of the PCEs series, box constraints and power flow equations are fulfilled with quite high accuracy. Future work will investigate the numerical scalability of ALADIN for larger grids and systematic parameter selection techniques. Moreover, we will investigate reserve capacity planning using the presented framework.

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